

PART-B

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INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-B of the examination has **4 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No questions require any clarification from the instructor or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	4	Total
Points:	6	7	5	2	20
Score:	6	7	5	2	20

QUESTIONS

1. (6 points) Let

$$S = \begin{pmatrix} S_{yy} & S_{yx} \\ (S_{yx})^T & S_{xx} \end{pmatrix}$$

be the partitioned sample covariance matrix of $Y_{1 \times 1}$ and $\mathbf{X}_{q \times 1}$ based on a centered (with respect to sample mean) sample of size n . Prove that the squared sample first canonical correlation r_1^2 is identical to the coefficient of determination (R^2) obtained from the multiple linear regression of Y on the vector $\mathbf{X}$.

- Considering: S_{yx} to be row ~~vector~~ matrix.
- We know that square of the first canonical correlation is the largest eigenvalue of: $\frac{S_{yx} S_{yx}'}{S_{xx} S_{yy}}$

$$S_{yy}^{-\frac{1}{2}} S_{yx} S_{xx}^{-1} (S_{yx})^T S_{yy}^{-\frac{1}{2}} \quad - \textcircled{1}$$

Since S_{yy} is $1 \times 1 \Rightarrow S_{yy}^{-\frac{1}{2}} = S_{yy}^{\frac{1}{2}} = \begin{bmatrix} \bullet \end{bmatrix}$
 $= \text{Var}(y)$ ~~$\times$~~

- We know that, in case of multiple linear regression of Y on vector X , the $\hat{\beta}$ vector is given by:

$$\hat{\beta} \leftarrow \text{estimate}$$

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

$$= (S_{xx})^{-1} (S_{yx})^T$$

$$4 \quad SS_{\text{Reg}} = (S_{yx}) (\hat{\beta})$$

$$= S_{yx} (S_{xx})^{-1} (S_{yx})^T$$

Now $R^2 = \frac{SS_{\text{Reg}}}{SS_T}$ in case of simple linear regression

~~$\therefore$ In multiple linear regression of Y on X
the $\frac{1}{SS_T}$ will be replaced by~~

~~⊗~~ Dimensions of ~~Z~~⊗:

$$\begin{matrix} S_{yy}^{-\frac{1}{2}} & S_{yx} & S_{xx}^{-1} & (S_{yx})^T & S_{yy}^{-\frac{1}{2}} & = & Z \\ 1 \times 1 & 1 \times q & q \times q & q \times 1 & 1 \times 1 & \end{matrix}$$

⇒ Net dimensions = 1×1

∴ When we try to find eigenvalue λ we do:

$$|Z - \lambda I| = 0$$

& since ~~Z has~~ is 1×1 say $[z_i]$

$$\Rightarrow |[z_i] - \lambda [1]| = 0$$

$$\Rightarrow z_i - \lambda = 0 \Rightarrow \lambda = z_i$$

∴ $r_i^2 = \lambda = z_i \Rightarrow$ Our r_i^2 is just the entry of 'Z'.

- In simple linear regression we had $R^2 = \frac{SS_{\text{Reg}}}{SS_T}$
- On extending it to multiple linear regression $\frac{SS_{\text{Reg}}}{SS_T}$ becomes same as dividing by variance of $Y \rightarrow$ because $SS_T = \sum (y_i - \bar{y})^2$ but we only have one entry same as variance which we know is the entry square of entry $\frac{1}{SS_T}$

⇒ We can conclude that R^2 is the value obtained by dividing the entry of

$$M = (S_{yx}) (S_{xx}^{-1}) (S_{yx})^T \text{ which is a } 1 \times 1 \text{ matrix}$$

by the var(X) which is exactly what $(S_{yy}^{-\frac{1}{2}}) \times M \times (S_{yy}^{-\frac{1}{2}})$ acts as in $Z \Rightarrow R^2$ is identical to

r_i^2

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Hence proven

2. Suppose you are performing a factor analysis on a 4×4 sample covariance matrix $\mathbf{S}$. The eigenvalues of $\mathbf{S}$ are $\lambda_1 = 0.8$, $\lambda_2 = 2.1$, $\lambda_3 = 4.2$, and $\lambda_4 = 0.2$. The corresponding normalized eigenvectors are $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, and $\mathbf{e}_4$. You decide to extract $m = 2$ factors using the principal component method.
- (2 points) Write down the explicit mathematical expression for the estimated factor loading matrix $\tilde{\mathbf{L}}$ in terms of the given eigenvalues and eigenvectors.
 - (1 point) Let the first row of $\tilde{\mathbf{L}}$ be $\tilde{\mathbf{l}}_1^T = [0.8, -0.4]$. Calculate the estimated communality $\tilde{h}_1^2$ for the first variable.
 - (4 points) Prove that the sum of all estimated communalities $\sum_{i=1}^p \tilde{h}_i^2$ is exactly equal to the sum of the largest $m = 2$ eigenvalues of $\mathbf{S}$.

• $\mathbf{S}_{k \times k} = \mu_{k \times k}^* + \tilde{\mathbf{L}}_{k \times m} \mathbf{F}_{m \times k} + \mathbf{E}_{k \times k}$

dash notation is just to differentiate from given λ 's.

a) $\tilde{\mathbf{L}} = \left[\sqrt{\lambda_1} \mathbf{e}_1^T \mid \sqrt{\lambda_2} \mathbf{e}_2^T \mid \sqrt{\lambda_3} \mathbf{e}_3^T \mid \sqrt{\lambda_4} \mathbf{e}_4^T \right]$

in order of largest eigenvalue first $\rightarrow$ only taking 2 because we have $m = 2$

$\tilde{\mathbf{L}} = \left[\sqrt{\lambda_3} \mathbf{e}_3^T \mid \sqrt{\lambda_2} \mathbf{e}_2^T \mid \sqrt{\lambda_1} \mathbf{e}_1^T \mid \sqrt{\lambda_4} \mathbf{e}_4^T \right]$

02 $\tilde{\mathbf{L}} = \left[\sqrt{4.2} \mathbf{e}_3^T \mid \sqrt{2.1} \mathbf{e}_2^T \mid \sqrt{0.8} \mathbf{e}_1^T \mid \sqrt{0.2} \mathbf{e}_4^T \right]$

$\approx \begin{bmatrix} \sqrt{4.2} & \sqrt{2.1} & \sqrt{0.8} & \sqrt{0.2} \end{bmatrix}$

b) $\tilde{h}_1^2 = \left(\frac{0.8}{\sqrt{4.2}} \right)^2 + \left(\frac{-0.4}{\sqrt{2.1}} \right)^2$

$\Rightarrow \tilde{h}_1^2 = \left(\frac{0.8}{\sqrt{4.2}} \right)^2 \times \sqrt{4.2} + \left(\frac{-0.4}{\sqrt{2.1}} \right)^2 \sqrt{2.1}$

$= 0.31228$

$\tilde{h}_1^2 = (0.8)^2 + (0.4)^2$

$= 0.64 + 0.16$

$= 0.8$

01

c) Our $\tilde{L}$ ~~looks like~~ has dimensions

$$\begin{bmatrix} & \\ & \\ & \\ & \end{bmatrix}_{p \times 2}$$

Currently $\tilde{L}$ has $\sqrt{4.2} e_3$ as first column & $\sqrt{2.1} e_2$ as second
 • in general it is the largest two eigenvalues & corresponding eigenvectors in decreasing order.

for each ~~row~~ variable i , ~~h_i^2~~ ~~square of~~

$$\tilde{h}_i^2 = \underbrace{(\text{entry } [i, 1])^2}_{\substack{\text{of} \\ \tilde{L}}} + (\text{entry } [i, 2])^2_{\substack{\text{of} \\ \tilde{L}}}$$

$$= ((\sqrt{4.2})^2 \times (e_3^i)^2) + ((\sqrt{2.1})^2 \times (e_2^i)^2)$$

where $\star$ 4.2 & 2.1 are the largest 2 eigenvalues & e_3 and e_2 are their corresponding eigenvectors with e_3^i & e_2^i being their i^{th} component

$$\begin{aligned} \text{So } \sum_{i=1}^p \tilde{h}_i^2 &= \sum_{i=1}^p (4.2 \times (e_3^i)^2) + \sum_{i=1}^p (2.1 \times (e_2^i)^2) \\ &= 4.2 \sum_{i=1}^p (e_3^i)^2 + 2.1 \sum_{i=1}^p (e_2^i)^2 \end{aligned}$$

Since we are given that all of e_1, e_2, e_3, e_4 are normalised eigenvectors

$$\Rightarrow \sum_{i=1}^p (e_3^i)^2 = 1 \quad \& \quad \sum_{i=1}^p (e_2^i)^2 = 1$$

$$\begin{aligned} \Rightarrow \sum_{i=1}^p \tilde{h}_i^2 &= 4.2 \times 1 + 2.1 \times 1 \\ &= 4.2 + 2.1 \\ &= 6.3 \end{aligned}$$

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i.e. sum of largest 2 eigen values

Hence proved

For general case replace 4.2 by largest eigenvalue, 2.1 by second largest eigenvalue & e_3, e_2 become their corresponding normalised eigenvectors.

3. (5 points) Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a distribution with mean $\mu \in \mathbb{R}$ and variance μ^2 . Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Derive a variance-stabilizing transformation $g(\bar{X}_n)$ such that the asymptotic variance of $g(\bar{X}_n)$ does not depend on μ .

- X_i ~~has~~ has mean μ & variance μ^2

Now $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \rightarrow$ mean μ , variance $\rightarrow \frac{\mu^2}{n}$

- From delta method, we know that the asymptotic variance of $g(\bar{X}_n)$ will be $[g'(\mu)]^2 \text{Var}(\bar{X})$
 $= (g'(\mu))^2 \times \frac{\mu^2}{n}$

- For this to be independent of μ :

$$(g'(\mu))^2 \times \frac{\mu^2}{n} = C \quad \left(\begin{array}{l} C \rightarrow \text{constant or can have } n \\ \text{in it} \\ \text{but not } \mu \end{array} \right)$$

$$\Rightarrow (g'(\mu))^2 = \frac{cn}{\mu^2}$$

$$\Rightarrow g'(\mu) = \frac{\sqrt{cn}}{\mu}$$

$$\Rightarrow dg(\mu) = \frac{\sqrt{cn}}{\mu} d\mu$$

$$\Rightarrow \int dg(\mu) = \int \frac{\sqrt{cn}}{\mu} d\mu$$

$$\Rightarrow g(\mu) = \sqrt{cn} \ln|\mu| + C' \rightarrow \text{constant of integration}$$

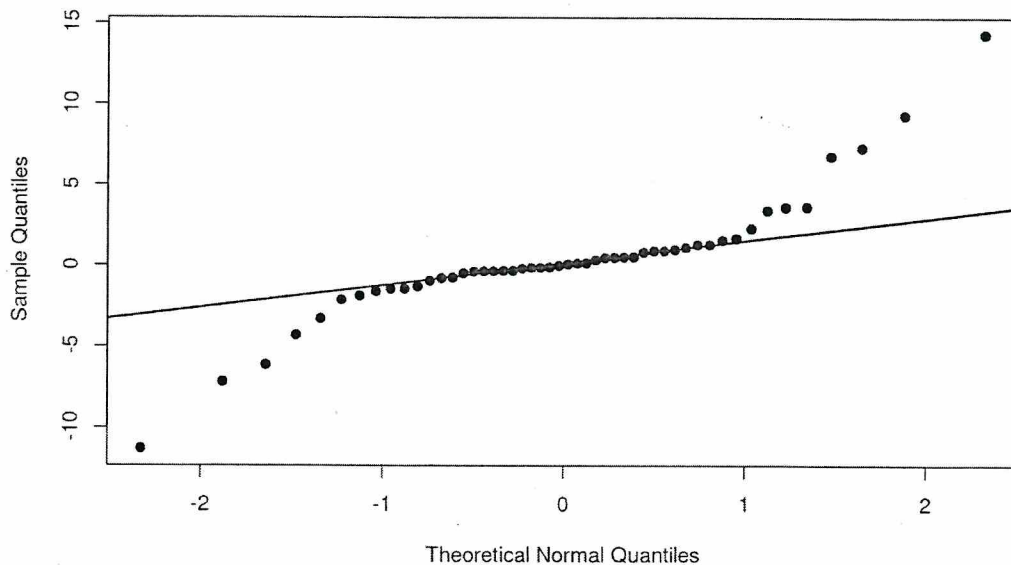
$$\therefore g(\bar{X}_n) = \sqrt{cn} \ln|\bar{X}_n| + C'$$

where c & c' are not functions of $(\bar{X}_n)$

$$g(\bar{X}_n) = \sqrt{cn} \ln|\bar{X}_n| + C'$$

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4. (2 points) The figure below displays a Normal Q-Q plot for a dataset of size $n = 50$. Interpret this plot to determine if the data follows a normal distribution. If the normality assumption is violated, specify which class of distributions would more accurately characterize the data.



- Since the data points do not lie on a straight line $\Rightarrow$ the quantiles of actual data ~~are~~ differ from quantiles of a normal distribution
 $\therefore$ The normality assumption is violated.
- ~~This~~ Cauchy distribution would more accurately describe this.

02

$$\phi^{-1}\left(\frac{i}{n+1}\right)$$

ADDITIONAL PAGES FOR ANSWERS

Answer the following questions in brief or as directed.
1. Define the term 'Business'.
2. List the characteristics of business.
3. Explain the difference between a business and a profession.
4. What are the types of business?
5. Define 'Entrepreneur'.

